Comparing Time to Recovery Between Initial and Repeat Concussion in Athletes

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Abstract

Objective: Compare time to recovery between initial and repeat concussions. **Design:** Retrospective review of electronic medical record. **Setting:** An interdisciplinary concussion clinic. **Patients:** Fifty patients (14.4 years old [±2.5]; 24 [48%] female) who sustained >1 concussion between 2016 and 2020 and were evaluated within 30 days of their concussion. **Intervention:** Injuries were divided into first, second, and/or last concussions. **Main Outcome Measures:** The primary outcome was time to recovery, as defined by clearance for return to sport. This was compared between first versus second concussion (primary analysis) and first versus last concussion (secondary analysis) using Cox regression. Multivariable models accounted for time from concussion to initial clinic visit and Sport Concussion Assessment Tool symptom severity score at initial encounter. Sensitivity analyses explored potential influences of initial encounter timing and interval between concussions. **Results:** Time to recovery did not differ between initial and repeat concussion in univariate or multivariable models (HR = 1.02, 95% CI, 0.65-1.62, $P = 0.92$; HR = 1.01, 95% CI, 0.64-1.62, $P = 0.91$ for multivariable models comparing first (ref) versus second and first (ref) versus last concussion, respectively). Similar results were obtained in sensitivity analyses stratifying by time to initial visit (≤7 days vs >7 days) or time between injuries (≤1 year vs >1 year). **Conclusions:** There were no differences in time to recovery between initial and repeat concussions sustained by the same athlete. A nonsignificant trend of shorter recovery after repeat concussions was observed for patients presenting within 1 week of injury and when repeat injuries occurred within 1 year.

Key Words: concussion, repeat concussion, time to recovery, SCAT symptom severity, time to initial visit, time between injuries

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INTRODUCTION

Concussion is a mild traumatic brain injury commonly sustained by athletes during sport participation. The cognitive, physical, and emotional symptoms associated with concussion can adversely affect daily functioning and limit a wide range of activities including athletics, academics, and work. Although the incidence of concussion is difficult to accurately surveil,[1,2] reports suggesting a recent increase in the incidence of sport-related concussion have led to the recognition of this injury as a growing public health concern in the United States.[3] Repeat concussions are common, with data from the National Collegiate Athletic Association (NCAA) suggesting that approximately 1 in 11 sports-related concussions are repeat injuries.[4] Concern surrounding repeat concussions has increased in light of emerging evidence suggesting a potential relationship between history of repetitive head impacts and/or multiple concussions and adverse effects on long-term neurologic health.[5–8] These relationships remain debated and additional high-quality studies are needed.[9]

A number of concussion risk factors have been identified, including history of concussion,[10] sport,[11,12] age,[13] and sex.[14] Of these, some studies have identified concussion history as the strongest single predictor for sustaining subsequent concussion,[10] although at least at the NCAA level, improvements in clinical management have helped mitigate this risk.[15] Additional risk factors have been identified as predictors of recovery duration, with symptom severity and preexisting mental health disorders being the strongest predictors in some studies.[16–18] Self-reported concussion history is also frequently associated with length of recovery although recall bias and potentially inaccurate self-reporting may confound this relationship.

Overall, there is mixed literature on the relationship between concussion history and recovery after subsequent concussion. Some studies have found longer recovery times in those with a self-reported concussion history,[19–21] while

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others report greater symptom severity without a longer recovery time.[22,23] Very few studies have examined recovery time after successive concussions in the same individual. Greater understanding of the effect of repeat concussions on recovery has potential to improve prognosis and management after concussion.[24–27]

The primary objective of this study was to compare recovery times after successive concussions sustained by the same individual. We evaluated time to recovery between athletes' first and second concussions, as well as their first and last concussions (for those treated for more than 2 concussions) managed at a single sports neurology clinic in an academic medical center. We also explored the potential influences of presentation time (ie, time from injury to initial clinic encounter) and time between concussions.

METHODS

Design, Setting, and Study Population

A retrospective analysis of electronic medical record (EMR) data was conducted across 521 patients who presented to a single interdisciplinary sports neurology clinic at an academic medical center between September, 2016, and February, 2020.[28] All patients were athletes with concussion whose goal was to return to sport, regardless of the mechanism of injury. Patients were eligible for inclusion if they were treated for more than 1 concussion, as diagnosed by a specialist physician based on the most recent consensus definition considering mechanism of injury, symptoms and symptom course, and neurologic examination findings.[24] The physicians making the concussion diagnosis were either fellowship-trained or attending neurologists and physiatrists with extensive experience treating athletes with concussion. Patients with initial clinic visits occurring more than 30 days postinjury were excluded because a large proportion of patients presenting to this clinic later than 1 month postinjury experience atypical recoveries with persisting postconcussion symptoms. Given the relative infrequency of repeat concussions in the same patient, no age-related exclusion criteria were applied. The study was approved by the institutional review board at the authors' university, which determined waiver of consent for this secondary analysis of EMR data.

Data Collection

Data were extracted from the EMR and compiled into a database including patient demographics and concussion history; days from injury to initial clinic visit; dates of injury and recovery for each concussion; injury setting and characteristics; and clinical findings at initial encounter, including self-reported symptoms on the Sport Concussion Assessment Tool (SCAT) symptom checklist.[24] Each individual injury was numbered as the patient's first, second, third, or fourth concussion treated in the clinic. Recovery date was defined based on physician written clearance for return to unrestricted sport participation. All physicians made recovery decisions based on return to preinjury baseline symptom levels, discontinuation of any rescue analgesic use, no concussion-related physical examination findings (including vestibular–ocular motor screening), and asymptomatic cognitive and physical activity. Asymptomatic physical activity included each patient undergoing in-clinic supervised exercise to greater than 80% of the predicted maximum heart rate without symptom provocation as done in Popovich et al.[29] Time to recovery was calculated for each injury by subtracting date of injury from recovery date.

Statistical Analysis

Descriptive statistics were calculated and paired *t*-tests and χ^2 tests were used to compare sex, race, and concussion history between first and second/last concussions. Univariate Cox proportional hazards regression was then used to compare time to recovery between first and second concussions (primary analysis) and between first and last concussions (secondary analysis). Observations with recovery times greater than 100 days were censored within the analysis. In the event patients were lost to follow-up before receiving clearance for return to sport or if they continued to receive treatment for symptoms not attributed to concussion, as determined by the treating physician, the patients were censored within the analyses at their last clinic visit when they had ongoing signs or symptoms of concussion and were known to be unrecovered (n = 26 injuries). The same data were used in primary and secondary analyses for 43 patients who were only treated in the clinic for 2 concussions, while different data were used for 7 patients who were treated for 3 or more injuries. Next bivariate relationships considering the influence of age, sex, time to initial clinic visit, time between injuries, SCAT symptom number and severity score, concussion history, and presence of loss of consciousness still considering concussion number (ie, first, second/last) were assessed using multivariable Cox proportional hazards regression. Final multivariable Cox proportional hazards regression models were then built to compare time to recovery between first and second/last concussions accounting for those covariates with significant bivariate effects (time to initial clinic visit and SCAT symptom severity score). Note that although SCAT symptom number and severity score both had significant bivariate effects, we elected to include only SCAT severity score in the multivariable models because of collinearity between these 2 variables (R = 0.33 for first vs second and R = 0.35 for first vs last concussions) and greater perceived clinical relevance. Both unadjusted and adjusted Kaplan–Meier curves were generated to visualize the relationships between initial and repeat injuries, with log-rank test *P*-values and median (±95% CI) time to recovery by injury number.

Given the critical role that initial visit timing plays in these analyses (eg, a patient cannot be cleared for return to sport/declared recovered on a date that precedes their initial clinic visit), additional sensitivity analyses were performed stratifying the study population by time to initial clinic visit in 2 ways. First, the population was stratified into patients with initial clinic visits occurring within 7 days of injury after both concussions (n = 19) and patients with initial clinic visits occurring after 7 days postinjury after both concussions (n = 13). Second, the population was stratified into patients with a 7-day-or-less difference in days from injury to initial clinic visit between both concussions (n = 28) and patients with a more than 7-day difference in days to initial visit between concussions (n = 22). Seven days was chosen to separate the facets approximately in half and was deemed a clinically relevant threshold because recovery times are often described in weeks. Also, given the potential that reinjury risk may be

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associated with time since concussion, another sensitivity analysis was performed stratifying the study population into patients with 1 year or less between concussions (n = 24) and patients with greater than 1 year between concussions (n = 26). Again, 1 year separated the facets approximately in half and was felt to be a clinically relevant threshold.

All statistical calculations were conducted with RStudio statistical software, version 2023.9.0 + 369.

RESULTS

Study Population

A total of 50 patients (48% female; mean age 14.4 (±2.5) years, range 7–24) treated for 105 concussions were included in this analysis. Most patients were treated for 2 concussions (n = 43, 86%), while 6 patients (12%) were treated for 3 concussions and 1 patient (2%) was treated for 4 concussions. About half of the patients (n = 26; 52%) reporting a concussion history before the first injury were included in this analysis. Injury setting is illustrated in Figure 1, with 39 (36.2%) injuries occurring in collision sports, 36 (34.3%) in contact sports, 7 (6.7%) occurring in noncontact sports, and 19 (18.1%) occurring outside of sport. Additional characteristics of the study population are presented in Table 1.

Characteristics of First and Repeat Concussions

Comparisons between first and subsequent (second and last) concussions are presented in Table 2. As expected, participants were significantly older at the time of subsequent concussions ($P = 0.009$ for first vs second, $P = 0.003$ for first vs last comparisons). First and subsequent concussions did not differ by time from injury to initial clinic visit, SCAT symptom number or severity score at initial visit, presence of loss of consciousness, or injury setting. Similarly, the proportion of patients experiencing a prolonged recovery (as defined by recovery time >30 or >90 days) did not differ between first and subsequent injuries.

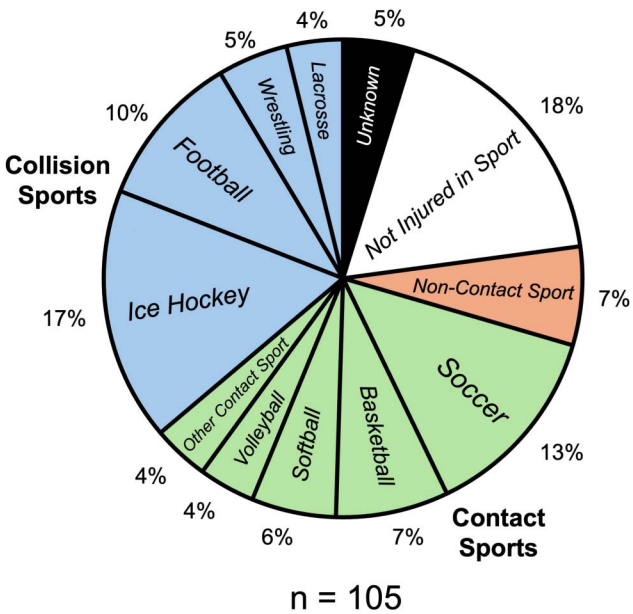

Figure 1. Injury setting across all 105 concussions analyzed. Noncontact sports included cheerleading, swimming, and biking (orange). Contact sports included soccer, basketball, softball, and volleyball (green). Other contact sports included field hockey (2%), diving (1%), gymnastics (0.5%), and motocross (0.5%). Collision sports included ice hockey, football, wrestling, and lacrosse (blue).

TABLE 1. Study Population Characteristics

Patient Demographics	n = 50
Age (yr)	14.4 (±2.5)
Sex (%)	
Male	26 (52)
Female	24 (48)
Race (%)	
White or Caucasian	39 (78)
Black or African American	4 (8)
Other	7 (14)
Number of concussions (%)	
2	43 (86)
3	6 (12)
4	1 (2)
Concussion history	26 (52)

Age is reported at the time of presentation after patients' first injury. Number of concussions reflects the number of injuries included in this analysis, while concussion history reflects self-reported injuries before the first included injury. Values are reported as mean (± SD) for continuous variables and frequency (%) for categorical variables.

Time to Recovery

Injury number was not associated with time to recovery when comparing first versus second (primary: HR = 1.39, $P = 0.008$) or first versus last (secondary: HR = 1.38, $P = 0.11$) concussions in univariate models (Figure 2). Median days until recovery after first, second, and last concussions were 42 days (95% CI, 30–48), 34 days (95% CI, 25–46), and 34 days (95% CI, 25–46), respectively. Independent relationships assessing the influence of each additional covariate on recovery time while accounting for injury number are presented in Table 3. Significant univariate effects were present for days from injury to initial clinic visit (HR = 0.95, $P = 0.009$, primary; HR = 0.95, $P = 0.008$, secondary) as well as SCAT symptom number and SCAT symptom severity score at initial visit (HR = 0.92, $P = 0.0002$ and HR = 0.98, $P = 0.0004$, respectively, primary; HR = 0.93, $P = 0.0001$ and HR = 0.98, $P = 0.0004$, respectively, secondary). Specifically, less time to initial visit and lower symptom numbers/scores were associated with faster recoveries. No significant independent effects were identified for age, sex, concussion history, or presence of loss of consciousness (all $P > 0.05$). Injury number remained a nonsignificant predictor of recovery time when comparing first versus second (primary: HR = 1.02, $P = 0.92$) and first versus last (secondary: HR = 1.02, $P = 0.92$) concussions, accounting for timing of initial visit and SCAT severity in multivariable models (Table 4, Figure 3). Both timing of initial visit and SCAT severity score at initial visit remained significant predictors in multivariable models comparing first versus second and first versus last injuries (all $P < 0.05$, Table 4).

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TABLE 2. Comparisons Between First and Repeat Concussions

	Initial Injury	Repeat Injury	
	First (n = 50)	Second (n = 50)	Last (n = 50)
Age (yr)	14.4 (2.25)	15.7 (2.4)*	15.8 (2.4)*
Time to initial clinic visit (d)	10.8 (8.6)	7.7 (5.9)	7.8 (5.9)
Time between injuries (d)	—	483 (325)	528.7 (343.4)
Initial SCAT symptom number	9.5 (6.6)	8.3 (7.02)	8.8 (7.1)
Initial SCAT severity score	21.8 (19.2)	20.8 (22.8)	22.3 (23.6)
Loss of consciousness (%)			
No	39 (78)	33 (66)	34 (68)
Yes	5 (10)	5 (10)	5 (10)
Unsure	6 (12)	12 (24)	11 (22)
Injury setting (%)			
Collision sport	17 (34)	16 (32)	18 (36)
Contact sport	19 (38)	15 (30)	17 (34)
Noncontact sport	5 (10)	8 (16)	4 (8)
Not injured in sport	9 (18)	11 (22)	11 (22)
Prolonged recovery (%)			
>30 Days	25 (50)	22 (44)	24 (48)
>90 Days	7 (14)	4 (8)	4 (8)

Values are reported as mean (± SD) for continuous variables and frequency (%) for categorical variables. Bolded variables are included as covariates in multivariable Cox regression models.
** Statistically significant ($P < 0.05$).*

Sensitivity analyses revealed that time to recovery did not differ between first and second concussions, nor between first and last concussions, regardless of whether the data were stratified by both initial clinic visits occurring greater than versus less than 7 days postinjury (see **Figure, Supplemental Digital Content 1**, http://links.lww.com/JSM/A484), a greater than versus less than 7-day difference in postinjury days at initial visit between the 2 concussions (see **Figure, Supplemental Digital Content 2**, http://links.lww.com/JSM/A485), or greater than versus less than 1-year between injuries (see **Figure, Supplemental Digital Content 3**, http://links.lww.com/JSM/A486) (all $P > 0.05$). The group of patients initially assessed within 7 days of injury recovered faster than patients not initially evaluated until later than 7 days postinjury, regardless of whether it was their first or repeat concussion (see **Figure, Supplemental Digital Content 1**, http://links.lww.com/JSM/A484, see **Table, Supplemental Digital Content 1**, http://links.lww.com/JSM/A487). Similarly, those patients who had a less than 7-day difference in time to initial clinic visit between their injuries recovered faster than those with a greater than 7-day difference, regardless of whether the injury was their first or subsequent (see **Figure, Supplemental Digital Content 1**, http://links.lww.com/JSM/A484, see **Table, Supplemental Digital Content 1**, http://links.lww.com/JSM/A487).

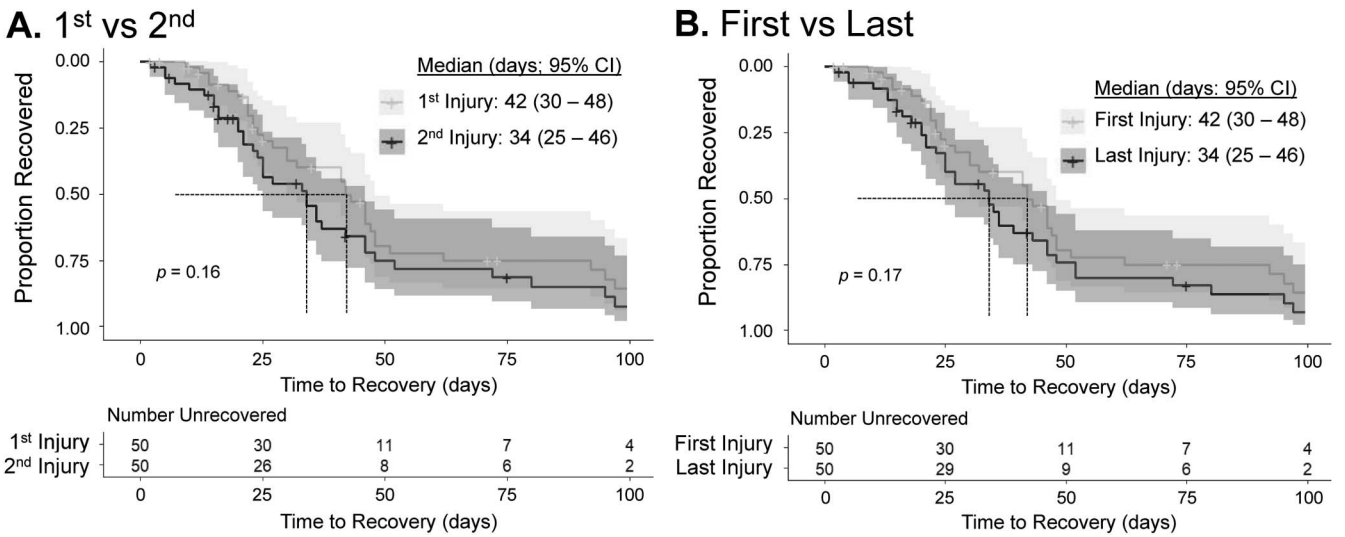

Figure 2. Unadjusted Kaplan–Meier survival curves showing days to recovery for first versus second injuries (A) and first versus last injuries (B). Log-rank tests *P*-values are presented and shaded regions indicate 95% confidence interval. "+" symbol indicates censored data.

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TABLE 3. Separate Bivariate Relationships to Recovery

Predictor (+ Injury Number)	First vs Second Injury			First vs Last injury		
	HR	95% CI	*P*	HR	95% CI	*P*
Sex						
Female	*Ref*	*Ref*	*Ref*	*Ref*	*Ref*	*Ref*
Male	1.39	0.80-2.42	0.24	1.37	0.80-2.32	0.25
Age	0.96	0.84-1.10	0.57	0.98	0.85-1.12	0.74
Time to initial clinic visit	0.95	0.91-0.99	0.009	0.95	0.91-0.99	0.008
Time between injuries	1.00	0.99-1.01	0.96	1.00	0.99-1.01	0.24
SCAT symptom number	0.92	0.89-0.96	0.0002	0.93	0.89-0.97	0.001
SCAT severity score	0.98	0.96-0.99	0.0004	0.98	0.96-0.99	0.004
Concussion history						
No	*Ref*	*Ref*	*Ref*	*Ref*	*Ref*	*Ref*
Yes	0.68	0.41-1.16	0.16	0.69	0.42-1.17	0.17
Loss of consciousness						
No	*Ref*	*Ref*	*Ref*	*Ref*	*Ref*	*Ref*
Yes	1.35	0.56-3.23	0.50	1.15	0.58-2.26	0.69
Unsure	1.05	0.58-1.90	0.88	0.97	0.52-1.80	0.92

Each predictor variable's relationship with the primary outcome variable—time to recovery—was examined with injury number using Cox regression analysis. This was conducted using first and second injury data as well as first and last injury data. P-values reflect log-rank test results. Bold indicates variables included as covariates in multivariable models.

DISCUSSION

In this study, concussion recovery was uniquely studied by comparing recovery times after initial and subsequent concussions sustained by the same patient, treated within the same clinical practice. We found no difference in time to recovery between initial and repeat concussions, regardless of whether patients' first versus second concussions or first versus last concussions were considered. Time to recovery after subsequent concussions actually tended to be slightly shorter than initial concussions, although not reaching statistical significance.

Prior research on this topic has yielded mixed results. Some studies have suggested a self-reported history of concussion is a predictor of prolonged recovery after subsequent concussion,[20,21] with a stronger relationship with history of 2 or more concussions.[19] Conversely, Gaudet et al.[30] recently found no differences in time until return to sport or return to school among high school athletes with versus without self-reported history of concussion. Another study more similar to our own conducted by Tang et al.[31] using a within-subject analysis adjusting for time to initial clinic visit and SCAT severity score at initial visit found no differences in the time until symptom resolution between initial and repeat concussions. Additional research by Preszler et al.[32] also using a within-subject analysis has similarly suggested that individuals may recover in less time and have reduced symptom burden after repeat concussions. Our own results align more closely with these within-subject analyses demonstrating no differences in recovery between initial and subsequent concussions in the same individual.

TABLE 4. Multivariable Time to Recovery Model

	First vs Second Injury			First vs Last injury		
	HR	95% CI	*P*	HR	95% CI	*P*
Injury						
Initial	*Ref*	*Ref*	*Ref*	*Ref*	*Ref*	*Ref*
Repeat	1.02	0.65-1.62	0.92	1.01	0.64-1.62	0.91
Time to initial clinic visit	0.95	0.91-0.99	0.01	0.94	0.90-0.99	0.008
SCAT severity score	0.98	0.96-0.99	0.0008	0.98	0.96-0.99	0.008

All of the variables were included in the same multivariable Cox regression analysis model with the primary outcome being time to recovery. P-values reflect log-rank test results.

Differences in how concussion history is defined, clinical management, and the outcome measures studied (often recovery and symptom sequelae), may contribute to these discrepant findings. Importantly, most previous studies have used a binary (yes/no) or numerical (eg 1, 2, 3+) designation for "history of previous concussion" based on self-report. These metrics are subject to potential recall bias as well as potential for inconsistent and/or unreliable reporting of prior undiagnosed concussions. Differences in clinical management across clinics and/or providers may also contribute to discordant findings. Despite consensus guidelines for return to sport and return to academics after concussion, management may differ across clinics and providers, especially for repeat concussions where providers may choose to be more conservative in their management and recommendations. Also, many studies use time until return to sport or return to academics as outcomes, both of which are subject to potential confounding factors such as frequency and timing of return visits across clinics. Symptom-based outcome measures such as time to asymptomatic rely on subjective symptom reporting and may also be subject to similar confounders. Our use of a within-subject design for patients treated within the same clinic accounts for many of these potential confounders.

This study accounted for 2 factors that are known to be associated with recovery in the multivariable analyses, specifically initial symptom severity and initial clinic visit timing. Multiple studies support higher symptom severity as

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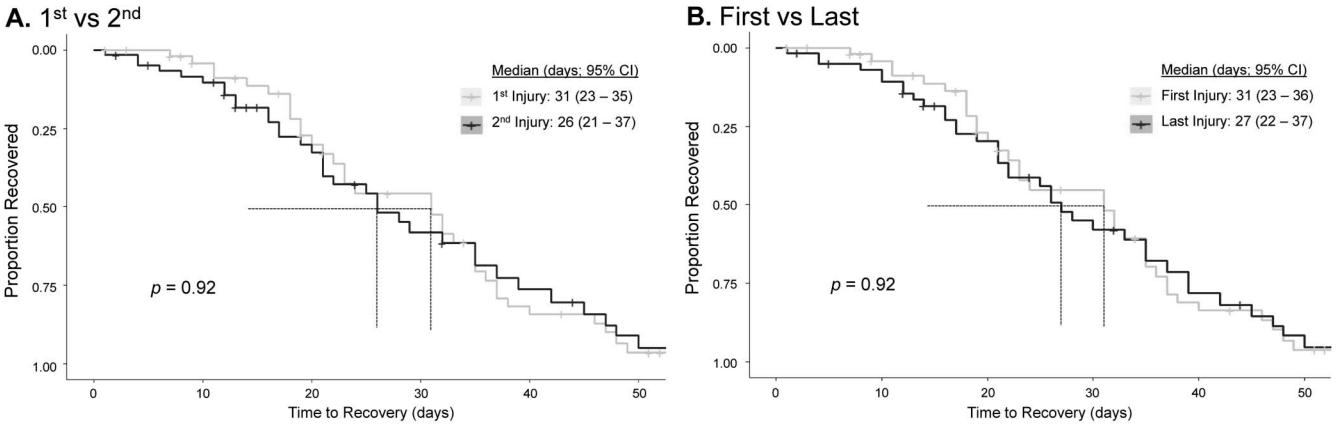

Figure 3. Adjusted Kaplan–Meier survival curves showing days to recovery for first versus second injuries (A) and first versus last injuries (B) accounting for time to initial clinic presentation and initial SCAT symptom severity scores. "+" symbol indicates censored data.

a predictor of longer recovery time,[16–18] consistent with our results. Also consistent with our results, existing literature supports faster recovery times in patients whose initial clinic assessments occur sooner after injury.[33–35] Moreover, our results reinforce the importance of symptom disclosure to promote earlier injury identification and treatment initiation. When accounting for initial symptom severity and timing of initial clinic visits in multivariable models, recovery time remained unrelated to initial versus subsequent concussion status, while both symptom severity and clinic visit timing remained significantly associated with recovery time. A potential contributor to slightly faster recovery after subsequent concussions may be that athletes who have previously sustained concussions and/or who are already familiar with a specialty concussion clinic, as well as their athletic trainers, may be more cognizant of repeat injuries and more inclined to seek care sooner thereby promoting a more rapid recovery.

Our results differed from previous work suggesting that younger age is associated with longer recovery time.[13] Although this study included patients across a wide age range overall, most were tightly distributed in middle adolescence, which may have mitigated potential age effects. Our results showed no effect of time between concussions on recovery, which also differs from some prior research suggesting that a shorter time interval between subsequent concussions is associated with a longer symptom duration after the repeat injury.[36] Most patients in this study sustained subsequent concussions many months after their initial injuries, which may have attenuated potential time interval effects. Furthermore, variability in provider care patterns may also potentially have played a role if other clinics tend to favor more conservative management of repeat concussions and/or wait longer to initiate early exercise or referrals to spine, vestibular, physical therapy, or mental health counseling than the providers caring for patients in this study.

Limitations

This study is not without limitations. The observational study design does not permit conclusions of causality. Furthermore, as a retrospective review of the EMR, the data were limited to what was documented through routine clinical care rather than a standardized research protocol. In addition, the inclusion criteria reduced the sample size heavily from the original source clinical population. Also, because only 7 patients sustained 3 or more concussions, there was substantial overlap between analyses of first versus second and first versus last concussions. In addition, as with much concussion recovery research in clinical populations, external factors such as timing of clinic visits may have potentially influenced recorded recovery dates. For example, patients might have recovered 1 or more days before their next clinic visit when their physician documented recovery and clearance for return to sport. Furthermore, it is also possible that follow-up clinic visits and, therefore, documentation of recovery may have been delayed by scheduling, transportation, other barriers, or that some patients who were recovering well and not in season (not requiring immediate written clearance for return to sport) may have elected to follow-up at a later time. In cases when patients were lost to follow-up or if their symptoms were deemed by the treating physician to no longer be attributable to concussion, the latest date they were known to be unrecovered was recorded and they were censored from analysis beyond that time. Also, this study took place for multiple years and included multiple providers, so it is possible that clinical management may have differed over time between providers. Patients presenting later to the clinic following their injury may represent a self-selected group who are already taking longer to recover, although our negative sensitivity analyses based on the timing of initial clinic visit do not suggest this possibility is likely to be influencing our results. Lastly, patients initially evaluated up to 30 days postinjury were eligible for inclusion, and there was likely a trade-off between symptom severity at initial visit and time until initial visit whereby patients seen more immediately postinjury likely had higher symptom burdens than patients not initially seen until later.

CONCLUSIONS

Within-subject analyses revealed no differences in time to recovery between initial and repeat concussions in analyses comparing first versus second or first versus last concussions and were not influenced by the effects of age or sex. These results do not support the hypothesis that recovery times are longer after subsequent concussions. Our findings can inform clinical treatment of patients with history of 1 or more prior concussions.

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References

1. Baldwin GT, Breiding MJ, Dawn Comstock R. Epidemiology of sports concussion in the United States. *Handb Clin Neurol*. 2018;158:63–74.
2. Pierpoint LA, Collins C. Epidemiology of sport-related concussion. *Clin Sports Med*. 2021;40:1–18.
3. Coronado VG, Haileyesus T, Cheng TA, et al. Trends in sports- and recreation-related traumatic brain injuries treated in US emergency departments: the national electronic injury surveillance system-all injury program (NEISS-AIP) 2001-2012. *J Head Trauma Rehabil*. 2015;30: 185–197.
4. Zuckerman SL, Kerr ZY, Yengo-Kahn A, et al. Epidemiology of sports-related concussion in NCAA athletes from 2009-2010 to 2013-2014: incidence, recurrence, and mechanisms. *Am J Sports Med*. 2015;43:2654–2662.
5. Manley G, Gardner AJ, Schneider KJ, et al. A systematic review of potential long-term effects of sport-related concussion. *Br J Sports Med*. 2017;51:969–977.
6. Gallo V, Motley K, Kemp SPT, et al. Concussion and long-term cognitive impairment among professional or elite sport-persons: a systematic review. *J Neurol Neurosurg Psychiatry*. 2020;91:455–468.
7. Morales JS, Valenzuela PL, Saco-Ledo G, et al. Mortality risk from neurodegenerative disease in sports associated with repetitive head impacts: preliminary findings from a systematic review and meta-analysis. *Sports Med*. 2022;52:835–846.
8. Yumul JN, McKinlay A. Do multiple concussions lead to cumulative cognitive deficits? A literature review. *PM R*. 2016;8:1097–1103.
9. Iverson GL, Castellani RJ, Cassidy JD, et al. Examining later-in-life health risks associated with sport-related concussion and repetitive head impacts: a systematic review of case-control and cohort studies. *Br J Sports Med*. 2023;57:810–821.
10. Abrahams S, Fie SM, Patricios J, et al. Risk factors for sports concussion: an evidence-based systematic review. *Br J Sports Med*. 2014;48:91–97.
11. Putukian M, Riegler K, Amalfe S, et al. Preinjury and postinjury factors that predict sports-related concussion and clinical recovery time. *Clin J Sport Med*. 2021;31:15–22.
12. Pfister T, Pfister K, Hagel B, et al. The incidence of concussion in youth sports: a systematic review and meta-analysis. *Br J Sports Med*. 2016;50: 292–297.
13. Foley C, Gregory A, Solomon G. Young age as a modifying factor in sports concussion management: what is the evidence?. *Curr Sports Med Rep*. 2014;13:390–394.
14. Berz K, Divine J, Foss KB, et al. Sex-specific differences in the severity of symptoms and recovery rate following sports-related concussion in young athletes. *Phys Sportsmed*. 2013;41:58–63.
15. McCrea M, Broglio S, McAllister T, et al. Return to play and risk of repeat concussion in collegiate football players: comparative analysis from the NCAA Concussion Study (1999-2001) and CARE Consortium (2014-2017). *Br J Sports Med*. 2020;54:102–109.
16. Iverson GL, Gardner AJ, Terry DP, et al. Predictors of clinical recovery from concussion: a systematic review. *Br J Sports Med*. 2017;51:941–948.
17. Kowalczyk CL, Eagle SR, Holland CL, et al. Average symptom severity and related predictors of prolonged recovery in pediatric patients with concussion. *Appl Neuropsychol Child*. 2022;11:145–149.
18. Miller JH, Gill C, Kuhn EN, et al. Predictors of delayed recovery following pediatric sports-related concussion: a case-control study. *J Neurosurg Pediatr*. 2016;17:491–496.
19. Thomas DJ, Coxe K, Li H, et al. Length of recovery from sports-related concussions in pediatric patients treated at concussion clinics. *Clin J Sport Med*. 2018;28:56–63.
20. Aggarwal SS, Ott SD, Padhye NS, et al. Clinical and demographic predictors of concussion resolution in adolescents: a retrospective study. *Appl Neuropsychol Child*. 2019;8:50–60.
21. Aggarwal SS, Ott SD, Padhye NS, et al. Sex, race, ADHD, and prior concussions as predictors of concussion recovery in adolescents. *Brain Inj*. 2020;34:809–817.
22. Mooney J, Pate J, Cummins I, et al. Effects of prior concussion on symptom severity and recovery time in acute youth concussion. *J Neurosurg Pediatr*. 2022;30:263–271.
23. Ellis M, Krisko C, Selci E, et al. Effect of concussion history on symptom burden and recovery following pediatric sports-related concussion. *J Neurosurg Pediatr*. 2018;21:401–408.
24. McCrory P, Meeuwisse W, Dvorak J, et al. Consensus statement on concussion in sport-the 5(th) international conference on concussion in sport held in Berlin, October 2016. *Br J Sports Med*. 2017;51:838–847.
25. McCrory P, Meeuwisse WH, Aubry M, et al. Consensus statement on concussion in sport: the 4th international conference on concussion in sport, Zurich, November 2012. *J Athl Train*. 2013;48:554–575.
26. Patricios JS, Schneider GM, van Ierssel J, et al. Beyond acute concussion assessment to office management: a systematic review informing the development of a Sport Concussion Office Assessment Tool (SCOAT6) for adults and children. *Br J Sports Med*. 2023;57:737–748.
27. Patricios JS, Schneider KJ, Dvorak J, et al. Consensus statement on concussion in sport: the 6th international conference on concussion in sport-Amsterdam, October 2022. *Br J Sports Med*. 2023;57:695–711.
28. Hanauer DA, Mei Q, Law J, et al. Supporting information retrieval from electronic health records: a report of University of Michigan's nine-year experience in developing and using the Electronic Medical Record Search Engine (EMERSE). *J Biomed Inform*. 2015;55:290–300.
29. Popovich M, Sas A, Almeida AA, et al. Symptom provocation during aerobic and dynamic supervised exercise challenges in adolescents with sport-related concussion. *J Athl Train*. 2021;56:148–156.
30. Gaudet CE, Iverson GL, Zafonte R, et al. Prior concussion history and clinical recovery after sport-related concussion in high school athletes. *J Neurotrauma*. 2023;40:1459–1469.
31. Tang AR, Williams KL, Davis PJ, et al. Symptom resolution following a repeat concussion within the same athlete. *Brain Inj*. 2024;38:295–303.
32. Preszler J, Manderino L, Eagle SR, et al. Evaluating recovery after two and three repeated concussions using growth curves. *J Neurotrauma*. 2024;41: 1409–1416.
33. Wingerson MJ, Magliato SN, Smulligan KL, et al. Predicting time to evaluation after pediatric concussion: factors affecting specialty concussion care. *Orthop J Sports Med*. 2023;11:23259671231186430.
34. Kontos AP, Jorgensen-Wagers K, Trbovich AM, et al. Association of time since injury to the first clinic visit with recovery following concussion. *JAMA Neurol*. 2020;77:435–440.
35. Eagle SR, Puligilla A, Fazio-Sumrok V, et al. Association of time to initial clinic visit with prolonged recovery in pediatric patients with concussion. *J Neurosurg Pediatr*. 2020;26:165–170.
36. Eisenberg MA, Andrea J, Meehan W, et al. Time interval between concussions and symptom duration. *Pediatrics*. 2013;132:8–17.

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